Decision Theory

Video Lesson

Duration: 00:06:04

Slide Deck

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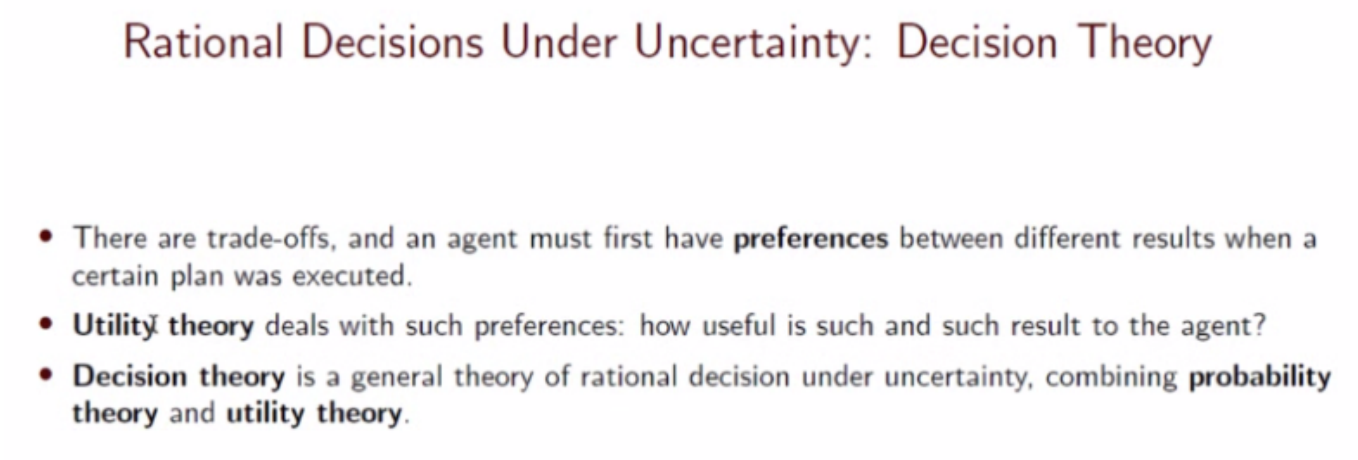
Lesson Transcript

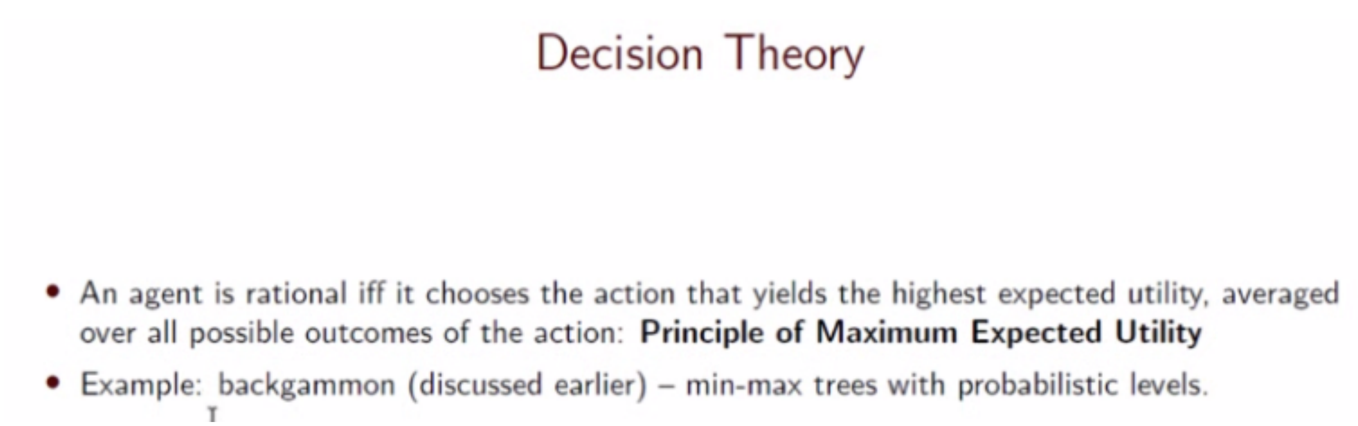
In this lecture, we will look into decision theory.



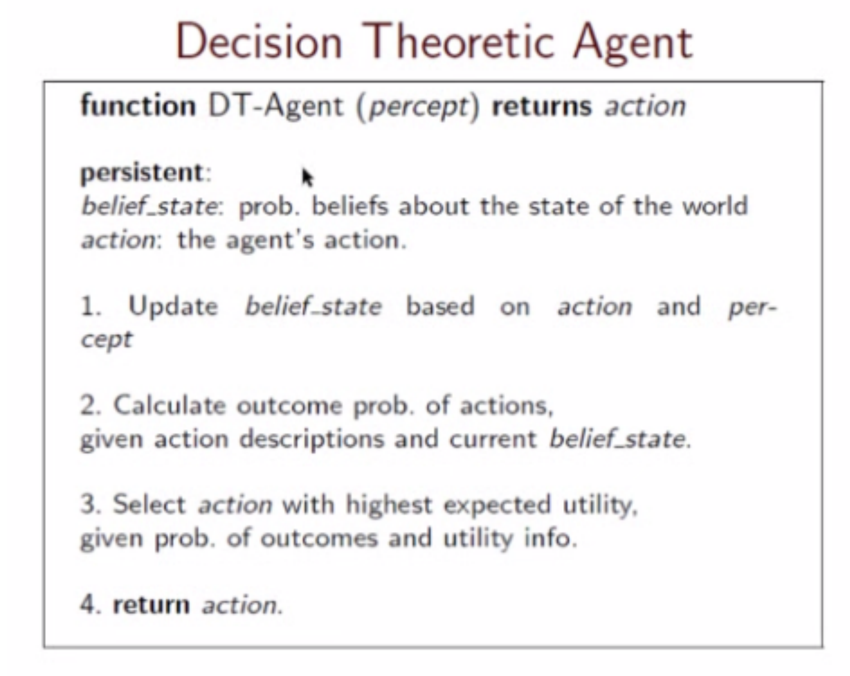
Decision theory tells us how to make rational decisions under uncertainty.

In decision theory, there are tradeoffs, and an agent must first have some kind of preference between different results when a certain plan is executed. The two main components of decision theory are utility theory and probability theory. Utility theory deals with these preferences. How useful is such and such result to the agent?

Decision theory is a general theory of rational decision under uncertainty that combines probability theory with utility theory.



Decision theory operates on the principle of maximum expected utility. So, an agent is rational if and only if it chooses the action that yields the highest expected utility averaged over all possible outcomes of the action. For example, in Backgammon, as we discussed earlier, we have this minimax tree with probabilistic levels to compute the expected utility.



Here is the decision-theoretic agent algorithm. So, it is a function that takes percepts. This is basically the observation, and based on that, it returns actions. So, these two, belief state and action, are persistent storage. So, the belief state will hold the probabilistic beliefs about the state of the world. An action would store the agent's action.

So, whenever you call this, the previous belief states and actions are made available to this function. First, you must update the belief state based on your action and percept. Then, calculate the outcome probability of these actions and give the action descriptions and current belief state.

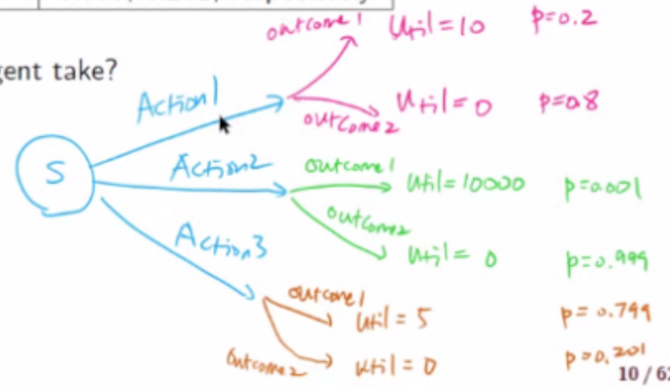
Next, you want to select the action with the highest expected utility, given the probability of outcomes and utility information that you updated. And once you select that action, you can return that.

Decision Theory: Example

Decision theory = Probability theory + Utility theory

	Utility of Resulting State	Probability
Action 1	10 or 0	0.2, 0.8, respectively
Action 2	10000 or 0	0.001, 0.999, respectively
Action 3	5 or 0	0.799, 0.201, respectively

Which action would an optimal Decision Theoretic Agent take?



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Let us consider this example. Suppose you are in a particular state, and let us say there are three available actions, Action 1, 2, and 3.

So, when you take Action 1, depending on the probability, you can end up in a situation with utility 10. And in the second possible outcome, you end up in a situation where the utility is 0, and note the probabilities.

That is the same for Action 2. So, when you take Action 2, there are two possible outcomes. One with a utility of 10,000 with a probability of 0.001. And another outcome that is possible has a utility of zero with probably 0.999.

And again, the same for Action 3. So, when you choose Action 3, there are two possible outcomes. One has the utility of 5, and one has the utility of 0, with each probability of 0.799 and 0.201.

If you just look at these values, obviously, Action 2 is the most attractive, but the problem is that the probability of that outcome is very low. Then maybe you want to pick Action 1 because 10 is greater than 5. But if you compare the probability for the utility of 10 for Action 1, where the probability is 0.2. And here you have the result of Action 3 with Outcome 1 and a utility of 5. But s has a higher probability.

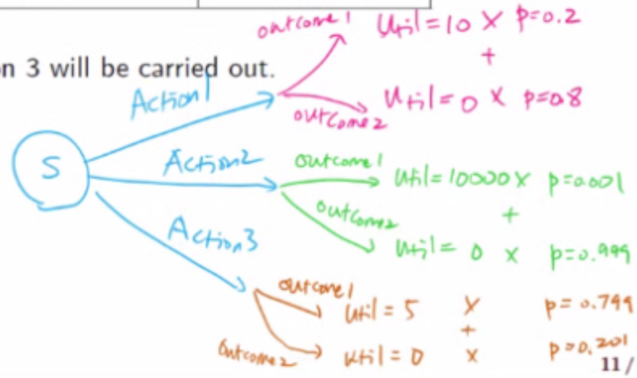
Now we want to make a decision. Which one of these actions to choose? So, decision theory, by combining probability theory and utility theory, will give you an optimal decision.

Decision Theory: Example

Decision theory = Probability theory + Utility theory

	$\sum$ Utility of Resulting State $\times$ Probability	Expected Utility
Action 1	$10 \times 0.2 + 0 \times 0.8$	2
Action 2	$1000 \times 0.001 + 0 \times 0.999$	1
Action 3	$5 \times 0.799 + 0 \times 0.201$	3.995 $\longleftarrow$

Action 3 has the maximum expected utility, thus action 3 will be carried out.



To compute the expected utility, which is the utility of the resulting state multiplied by that particular outcome's probability and summing up, we can get the expected utility for Action 1.

So, this is basically 10 times 0.2 plus zero times 0.8. That comes up to 2. Doing the same for Action 2 and Action 3, we get the expected value of 1 and 3.995.

Then we decide which action to take based on the action that gives us the highest expected utility. In this case, it will be Action 3.

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Action 3	$5 \times 0.799 + 0 \times 0.201$	3.995 $\longleftarrow$

Action 3 has the maximum expected utility, thus action 3 will be carried out.

In the following lectures, we look at probability basics and the Bayes rule.